ENERGY DETECTIVE: CRACK THE VAULT!

Squad Scorecard & Combination Tracker · Unit 1 Review Game

Detective Squad Name: _____ Detectives:

_____

CATEGORY	$100	$200	$300	$400	$500	DIGIT UNLOCKED
Case 1 Energy Causes Change	☐	☐	☐	☐	☐	Digit #1: ___
Case 2 Speed & Energy of Motion	☐	☐	☐	☐	☐	Digit #2: ___
Case 3 Transfer or Transformation	☐	☐	☐	☐	☐	Digit #3: ___
Case 4 Collisions	☐	☐	☐	☐	☐	Digit #4: ___
Case 5 Engineering & Edison/Latimer	☐	☐	☐	☐	☐	Digit #5: ___

$300 = Team Challenge · $500 = Team Explain · Each correct clue earns the points AND unlocks that case's digit.

MASTER VAULT COMBINATION

Digit 1	Digit 2	Digit 3	Digit 4	Digit 5
_____	_____	_____	_____	_____

TOTAL SCORE: _____ pts

TEACHER CARDS · Unit 1 Review Game

Cut these out. Team Explain ($500) = short spoken answer · Team Challenge ($300) = 30-second demo.

TEAM EXPLAIN CARDS ($500 CLUES)

Teams discuss, then explain in one or two spoken sentences. No long writing.

Case 1 · Energy Causes Change — TEAM EXPLAIN

A battery, a stretched rubber band, and a raised ball are not moving. Explain how they can still have energy.

Expected: *They have stored energy that can cause a change later.*

Case 2 · Speed & Energy of Motion — TEAM EXPLAIN

The same marble moved the cup farther from a higher ramp. Explain what this shows about speed and energy of motion.

Expected: *The faster marble had more energy of motion (a greater change).*

Case 3 · Transfer or Transformation — TEAM EXPLAIN

A flashlight lights a card across the room. Explain whether energy transferred, transformed, or both.

Expected: *Both — stored energy changes into light (transformation), and the light moves to the card (transfer).*

Case 4 · Collisions — TEAM EXPLAIN

A rolling car stops when it hits a target, and the target moves. Explain where the energy went.

Expected: *It transferred to the target (and some became sound). It did not disappear.*

Case 5 · Engineering & Edison/Latimer — TEAM EXPLAIN

Edison's first filaments burned out fast. Explain why testing more than 6,000 materials was useful.

Expected: *Each test was evidence that helped him improve the design.*

TEAM CHALLENGE CARDS ($300 clues)

30-second team demos. Act it out, then explain in one sentence.

Case 1 · Energy Causes Change — TEAM CHALLENGE

Stored Energy Freeze — freeze and show three examples of stored energy (a held book, a stretched rubber band, a battery).

Expected: *Each object is ready to cause a change later.*

Case 2 · Speed & Energy of Motion — TEAM CHALLENGE

Human Ramp — use your arms to model a low ramp and a high ramp. Show how ramp height changes speed and impact distance.

Expected: *A higher ramp means faster motion and a bigger change.*

Case 3 · Transfer or Transformation — TEAM CHALLENGE

Sound Transfer — model how sound moves energy across space without touching anything (vibrate to move a paper strip).

Expected: *The paper moving shows energy was transferred.*

Case 4 · Collisions — TEAM CHALLENGE

Collision Acting — act out a fast collision. Show motion transfer and sound.

Expected: *The moving object's energy transfers, and some becomes sound.*

Case 5 · Engineering & Edison/Latimer — TEAM CHALLENGE

Design Cycle Freeze — act out the design cycle: Problem → Design → Build → Test → Improve.

Expected: *Test results, even failures, are evidence to improve.*